FIN42110 - Financial Data Science

Detecting Informed Trading on Polymarket: A Geopolitical Prediction Market Analysis

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UCD Michael Smurfit Graduate Business School

Polymarket

johnstew324/polymarket-whale-watch

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0.1 Data Source Evaluation

Polymarket is a decentralised prediction market built on Polygon, a proof-of-stake Ethereum sidechain [Polymarket(2025)]. Users can trade on the outcomes of real-world events using USDC as collateral. The technical structure of these markets is defined by:

- **Market Mechanics:** Each market generates a pair of binary outcome tokens `YES`/`NO`. These outcomes are priced between $0 and $1, which directly reflects the market's implied probability of these event occurring.
- **Token Standards:** Each binary outcome is implemented as **ERC-1155** assets - fungible tokens with 6-decimal precision, traded in discrete units.
- **Settlement and Transparency:** All trades are settled through the CTF Exchange smart contract and are permanently recorded on-chain as `OrderFilled` events.
- **Data Accessibility:** both counterparty wallet addresses are therefore publicly visible in every event, making the full trade history queryable directly from the blockchain.

Trade History Acquisition:

Obtaining the history at wallet level and at scale required evaluating many sources of data. Table 1 summarises the sources evaluated and the reason each was rejected.

Table 1: Data sources evaluated and rejected.

Source	Reason rejected
Polymarket Data API `/data/trades`	Current live data but hard capped at 3,100 trades per market. 426 markets exceeded this threshold. These are the highest-volume geopolitical markets, precisely the most relevant for analysis.
Dune Analytics	Full history available in principle but free tier capped at ∼400,000 rows with complete access paywalled.
Allium `predictions.allium.so`	Not publicly accessible, requires enterprise demo approval.
CryptoHouse `clickhouse.com`	No Polymarket data, Ethereum and Solana only.
Goldsky `polymarket-orderbook-resync/prod`	A new subgraph that is actively indexed but does not include maker and taker wallet addresses entirely.

Chosen source: S3 snapshot of on-chain events. The complete `OrderFilled` event history was sourced from a community-maintained S3 archive published by `warproxxx/poly_data` [warproxxx(2025)]. The pre-built snapshot was used in place of scraping the live subgraph directly. Scraping the 2023-2025 window from scratch via GraphQL pagination takes over two days (as documented in the github repository). Validating that the modeling approach gives meaningful results before investing

in a full live pipeline, was the best approach. Extending the dataset beyond October 2025 and creating a live pipeline is a possible future improvement. Table 2 summarises the resulting dataset.

Table 2: S3 snapshot properties.

Property	**Value**
File	`orderFilled_complete.csv.xz`
Compressed size	5.9 GB
Uncompressed size	∼35–40 GB
Total rows	150M+
Date range	1 February 2023 - 7 October 2025
Hosted at	`polydata-archive.s3.us-east-1.amazonaws.com`

0.2 Pipeline Architecture

The data collection pipeline consists of a snapshot download followed by five Python modules executed sequentially, with each step's output feeding directly into the next. Figure 1 shows the full flow from the S3 archive and external APIs through to the analytical DuckDB database. Full setup and reproduction instructions are provided in the project `README.md`[1]

[1] johnstew324/polymarket-whale-watch

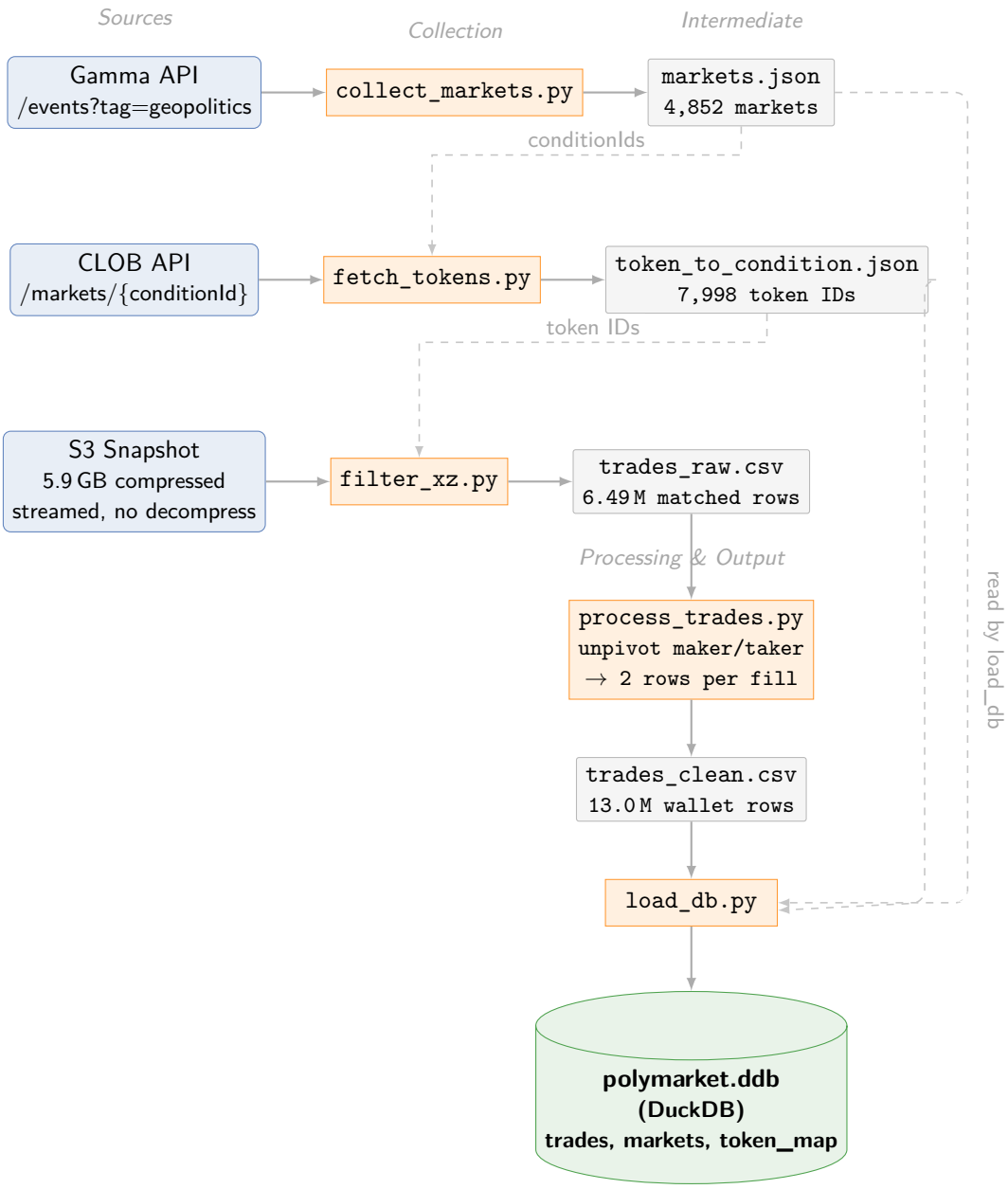

Figure 1: Data collection and loading pipeline. Orange: Python scripts (`src/pipeline/`), blue: external data sources, grey: intermediate files, green: analytical database. Solid arrows show the primary data flow; dashed arrows show secondary dependencies (metadata lookups). `filter_xz.py` streams the S3 snapshot without full decompression, matching token IDs on the fly. Full pipeline reproducible via `make pipeline`.

Step 0: S3 Snapshot Download:

The first step is downloading `orderFilled_complete.csv.xz` into `data/raw/` from the `warproxxx/poly_data` S3 archive (see Appendix A).

Table 3: Example rows from `orderFilled_complete.csv.xz`. Each trade assets are identified by ERC-1155 token IDs, these are 77-digit integers assigned at market creation, one for both Yes and No.`0` denotes USDC. No condition IDs or market names are present in the raw data, therefore the following steps of the pipeline exist to resolve this problem.

timestamp	maker	makerAssetId	makerAmt	takerAssetId	takerAmt
1675247746	`0x78db...`	`0` (USDC)	29.68	`55628...`	30.60
1675247746	`0xc6fd...`	`41572...`	30.60	`0` (USDC)	0.92

Step 1: `collect_markets.py`

The pipeline begins by querying the Gamma API for all closed geopolitical markets, paginating until exhausted, and writes the result to `data/raw/markets.json`. This file serves as the master market registry all over steps.

Listing 1: Gamma API query parameters.

```
r = httpx.get(f"{BASE}/events", params={ "limit": limit,
                                         "closed": "true",
                                         "tag_slug": "geopolitics",
                                         "offset": offset }) #
                                            filters: closed markets
                                             only, tagged with "
                                            geopolitics"
```

`resolvedOutcome` is not returned directly by the API, it is derived by parsing the `outcomePrices`, string into a list and selecting the outcome where the price goes to $1 at settlement. Over 5000 markets were retrieved with these parameters as of April 2026. The following table summarises the markets.json schema:

Table 4: `markets.json` field reference.

Field	Type	Example
`conditionId`	string	`0x0f99...`
`question`	string	Will Putin remain President through 2024?
`eventTitle`	string	Will Putin remain President through 2024?
`startDate`	ISO 8601	`2024-03-26T15:11:43Z`
`endDate`	ISO 8601	`2024-12-31T12:00:00Z`
`closedTime`	ISO 8601	`2025-01-01T09:42:22Z`
`volume`	float	`2,187,634`
`outcomes`	string[]	`["Yes", "No"]`
`outcomePrices`	string[]	`["1", "0"]`
`resolvedOutcome`	string	`Yes` (derived from `outcomePrices`)
`tags`	string[]	`["russia", "putin", "geopolitics", ...]`

Step 2 `fetch_tokens.py`

As shown in Table 3, snapshot rows carry no conditionID (cid), only ERC-1155 token IDs. `fetch_tokens.py` resolves this by querying the CLOB API for the YES/NO ERC-1155 token IDs of each market in `markets.json` with volume $\geq$\$1,000.

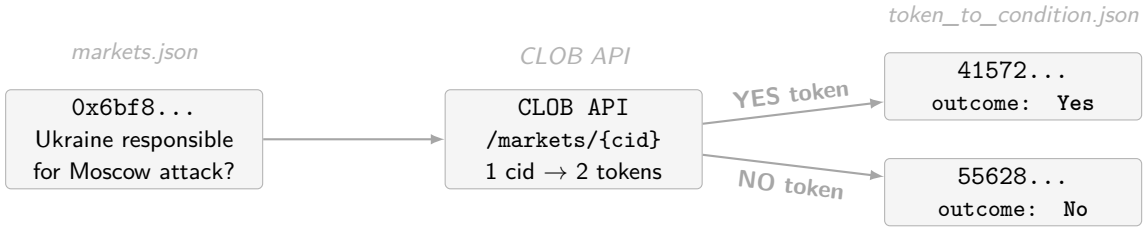

Figure 2: Each `conditionId` is resolved to a YES and NO ERC-1155 token ID via the CLOB API. The token IDs shown are real values from the dataset.

The \$1,000 volume filter excludes negligible markets, at time of writing, `token_to_condition.json` contained approximately 8,000 token IDs across 4,000 markets.

Step 3: `filter_xz.py`

Using the token to condition mapping, the `filter_xz.py` extracts matching trades from the full snapshot in a single streaming pass. Full decompression is impractical, given the archive size at 35-40 GB (Table 2). `xzcat`[2] (Unix utility) is used to stream the `.xz` file line by line into a Python CSV reader, without having to write the full decompressed file to disk.

Listing 2: Streaming the `.xz` archive without full decompression.

```
with subprocess.Popen(["xzcat", str(IN)],
                      stdout=subprocess.PIPE,
                      bufsize=1024*1024) as proc:
    reader = csv.reader(line.decode() for line in proc.stdout)
```

Each row is checked against `token_to_condition.json`: a match on either `makerAssetId` or `takerAssetId` means the row is written to the .csv file, with `condition_id` and `outcomes` appended to that output row.

Table 5: `trades_raw.csv` - output of `filter_xz.py`.

Property	Value
Rows	6,505,297 ($\approx$ 4.3% of full snapshot)
Unique wallets	333,124
File size	2.1 GB
Date range	1 February 2023 - 7 October 2025
Added columns	`condition_id`, `outcomes`

Step 4 `process_trades.py`

Each raw fill in `trades_raw.csv` records both the maker and taker in a single row. `process_trades.py` normalises this into wallet-centric rows suitable analysis:

- **Unpivot:** Each fill is split into two rows, a taker and a maker since every trade has exactly one buyer and seller. This expends the 6.5 M fills to 13.0 M wallet-centric rows.
- **Side inference:** For each `OrderFilled` event, the maker places the limit order and the taker fills it. When `makerAssetId` = 0 this indicates the maker paid USDC to acquire outcome tokens (`BUY`). Otherwise the maker delivered outcome tokens to receive USDC (`SELL`).
- **Normalisation:** raw integer amounts are divided by 10^6 to convert from ERC-1155 precision to USDC and token units
- **Price:** derived as `price` = `usd_amount`/`token_amount`, representing the implied probability at time of trade.

[2] https://linux.die.net/man/1/xzcat

- **Dust filter:** rows where the normalised token amount rounds to zero are discarded completely.

Table 6: Sample rows from `trades_clean.csv`. Each original fill becomes two rows with opposite sides.

timestamp	wallet	side	outcome	usd	tokens	price
2023-02-01	`0x78db...`	BUY	No	29.68	30.60	0.97
2023-02-01	`0x4bfb...`	SELL	No	29.68	30.60	0.97
2023-02-01	`0xc6fd...`	BUY	Yes	0.92	30.60	0.03
2023-02-01	`0x78db...`	SELL	Yes	0.92	30.60	0.03

Step 5: `load_db.py`

`load_db.py` is the final step of the core pipeline, loading the `trades_clean.csv` and `markets.json` into a single DuckDB analytical database (`polymarket.ddb`). Unix timestamps are converted into native `TIMESTAMP` columns, the `tags` field is converted from a raw JSON string to a native `VARCHAR[]` array. The token map is kept as a `token_map` table for possible future cross-reference. The resulting schema, and the database choice of DuckDB, are covered in Section 0.3.

0.3 Database Schema

All pipeline outputs are merged into a single DuckDB database (`polymarket.ddb`). DuckDB was chosen for three main reasons [Raasveldt and Mühleisen(2019)]:

- **Columnar execution:** its vectorised engine handles time-series aggregations and `GROUP BY` queries more efficiently than a row-oriented store such as SQLite, especially across 13 million wallet rows.
- **Native array support:** the `VARCHAR[]` type allows market tags to be stored and filtered directly, removing the need for a separate junction table.
- **Serverless:** like SQLite, no background process is needed, keeping the database fully portable as a single file within the project repository.

The database contains five tables, summarised in Table 7. Three are loaded by the core pipeline (`load_db.py`): `trades`, `markets`, and `token_map`. Two further tables, `sentiment` and `financial` are both populated by the sentiment pipeline (Section 1) and joined to the core tables by the shared foreign key, `condition_id`.

Table 7: Database tables after the full pipeline.

Table	Rows	Loaded by
`trades`	13,010,594	`load_db.py`
`markets`	5,372	`load_db.py`
`token_map`	8,794	`load_db.py`
`sentiment`	—	sentiment pipeline (Section 1)
`financial`	—	sentiment pipeline (Section 1)

Figure 3 shows the full schema and relationships between tables. `condition_id` is the shared foreign key that links trade activity, market metadata, token identifiers, and the sentiment and financial enrichment layers.

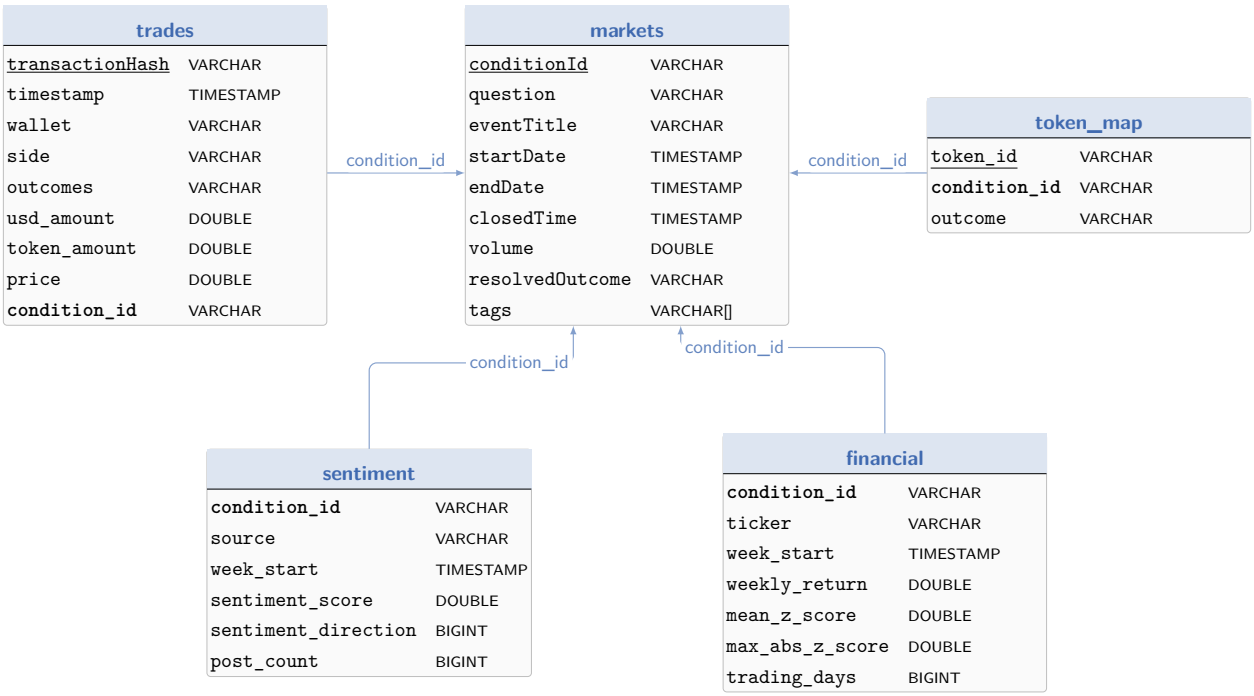

Figure 3: Database schema. Underlined fields denote primary keys; bold fields are foreign keys referencing `markets.conditionId`. The `sentiment` and `financial` tables are populated by the sentiment pipeline (Section 1).

An exploratory analysis of the resulting dataset, covering wallet activity distributions, volume concentration, hit rates, and market coverage-is provided in Section **??**.

1 Sentiment and Financial Signal Pipeline

The trade data from Section 0.2 records wallet activity, but not the information environment in which each trade was placed. A trade viewed completely in isolation is ambiguous, a YES position on a resolved market could reflect anything from luck to a reasonable read of public news to possible access to information not yet reflected in price. To attempt to separate these cases, each trade must be set against the external signals available during the same week.

Table 8 illustrates this with a real market. On 3 April 2025, a wallet buys YES on an Iran-Israel strike market at $0.12, against an information environment that was broadly sceptical. The strike occurs ten days later and the position resolves at $1.00. Without the surrounding context, the trade is indistinguishable from a lucky bet but with some extra context, the wallets deviation from the public consensus become more measurable.

Table 8: A single trade against its weekly information environment. The wallet enters at $0.12 during a week when all three external signals lean against the YES outcome; the market resolves YES ten days later. Signal values are illustrative.

Signal	**Score**	**Volume**	**Interpretation**
FT / FinBERT	−0.42	8 articles	Coverage emphasises diplomatic de-escalation
Truth Social / VADER	+0.05	2 posts	Neutral; no direct reference to Iran
IDF defence basket	−0.30	—	No abnormal move
Trade: wallet `0x78db...` buys YES at $0.12 on 2025-04-03 (week of 2025-03-31 to 2025-04-06)			
Resolution: YES settles at $1.00 on 2025-04-13			

This is the pattern the enrichment layer exists to surface: a wallet taking a position against these signals and being proven right. This looked at in more depth in Section **??**.

The remainder of this section describes the three sources. Each is aggregated to weekly resolution and keyed on `condition_id` + `week_start`:

- **Elite financial discourse.** Financial Times articles scored with FinBERT [**?**].
- **Executive political discourse.** Donald Trump's Truth Social posts scored with VADER [**?**], sourced from `stiles/trump-truth-social-archive` [**?**].
- **Market-implied consensus.** yfinance daily OHLCV for geopolitically-linked tickers, converted to weekly abnormal returns against a 20-day rolling baseline, with CBOE VIX as a global volatility benchmark.

Signals are joined at the week a wallet enters a market, not at resolution date. A bet with possible extra information might be placed months before the market closes, if the signals are only looked up at the resolution date, the rest of the market's life and the wallet's position would not be counted for.

1.1 Architecture Overview

Figure 4 shows the three parallel tracks that produce the enrichment tables. All three consume the shared `markets` table and NER module, and all three write into the same `polymarket.ddb` database established in Section 0.2. A single vocabulary module, `pipeline_config.py`, drives keyword-to-corpus, keyword-to-ticker, and author-stopword mappings across all three collectors — covered in the subsections that follow.

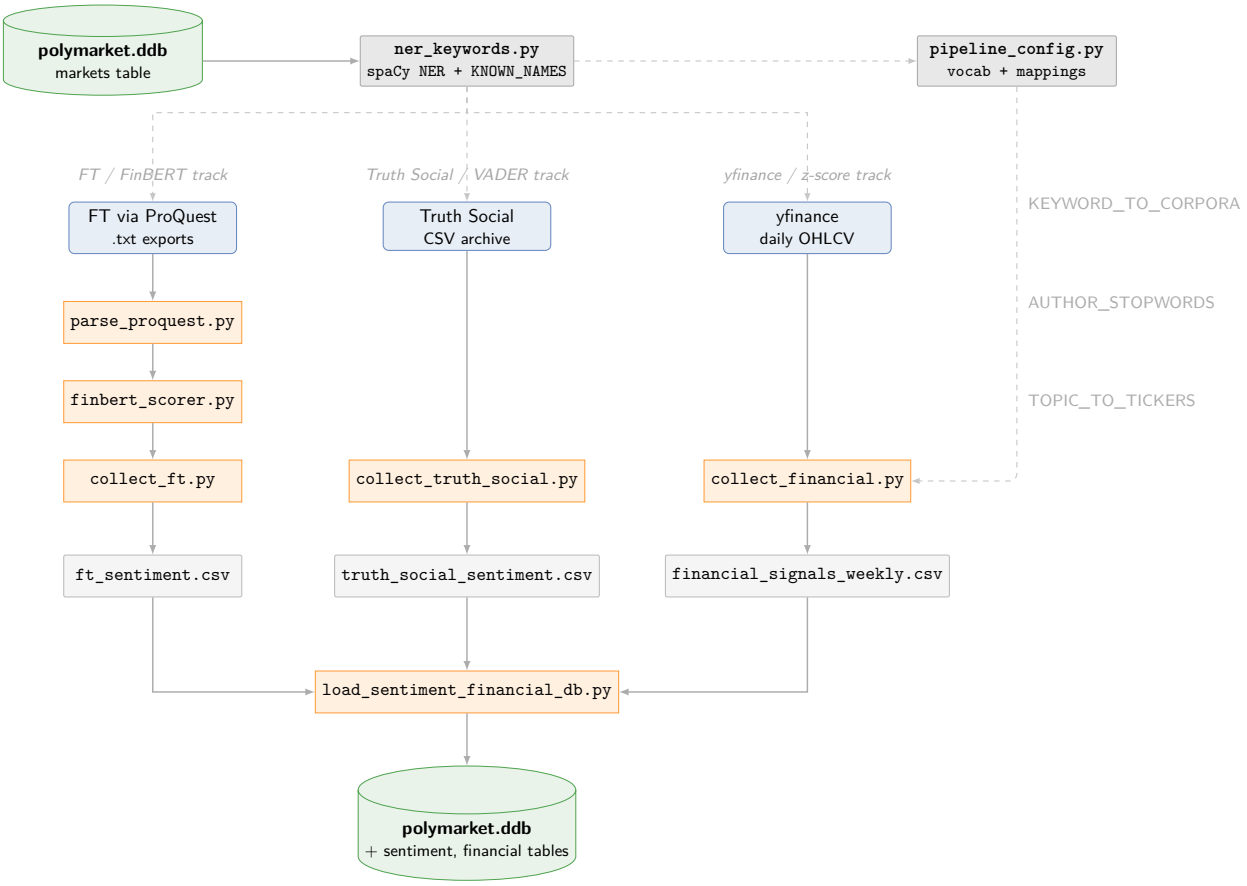

Figure 4: Sentiment and financial enrichment pipeline. Orange: Python scripts, blue: external data sources, grey: intermediate files and shared resources, green: DuckDB. Solid arrows: primary data flow; dashed arrows: vocabulary and configuration dependencies.

spaCy's default NER is not sufficient on its own for this domain. Market questions present three specific challenges:

- **Short and fragmented:** Questions are typically under 15 words and often start with proper nouns, confusing `spaCy`'s sentence parser.
- **Out-of-vocabulary entities:** Names outside `spaCy's` default English model include frontline town names (`Pokrovsk`, `Sudzha`, `Siversk`, `Kupiansk`), faction names (`Houthi`, `HTS`, `PKK`), and transliterated leaders (`Al-Sharaa`, `Khamenei`).

- **Surface-form variation:** The same entity appears in multiple forms such as `Benjamin Netanyahu`, `Netanyahu`, `Israel's PM`, `Bibi Netanyahu`. Which all need to be collapsed to a single canonical name.

`ner_keywords.py` handles these with a three-pass design backed by a shared vocabulary.

Three-Pass Extraction

The extractor loads spaCy's `en_core_web_sm` model and runs three passes over each question, each covering a failure mode of the previous one:

1. **spaCy NER:** Extract entities of type `GPE`, `PERSON`, `ORG`, or `NORP` and normalises it through the `KNOWN_NAMES`. Before parsing, a neutral prefix `"About:  "` is taken from the question so that sentence-initial proper nouns (`"Xi Jinping out in 2025?"`) are not misread as verbs.
2. **Proper noun fallback:** Any word tagged as a proper noun that Pass 1 missed is added and normalised. Catches names spaCy recognised as a name but did not classify.
3. **`KNOWN_NAMES` phrase lookup:** The remaining tokens are checked against `KNOWN_NAMES` in phrases of decreasing length. Checks three words first, then two, then one, so that multi-word names like `"Gulf Leaders Summit"` are matched as a single unit rather than split into separate keywords. This ultimately catches names spaCy does not recognise at all.

Listing 3: Pass 3: sliding phrase lookup in `KNOWN_NAMES`.

```
for i, token in enumerate(tokens):
    for n in (3, 2, 1):
        if i + n > len(tokens):
            continue
        phrase = " ".join(t.text for t in tokens[i:i + n]).lower()
        canonical = KNOWN_NAMES.get(phrase)
        if canonical and canonical.lower() not in seen:
            keywords.append(canonical)
            break
```

Figure 5 walks through a worked example and shows where each pass contributes.

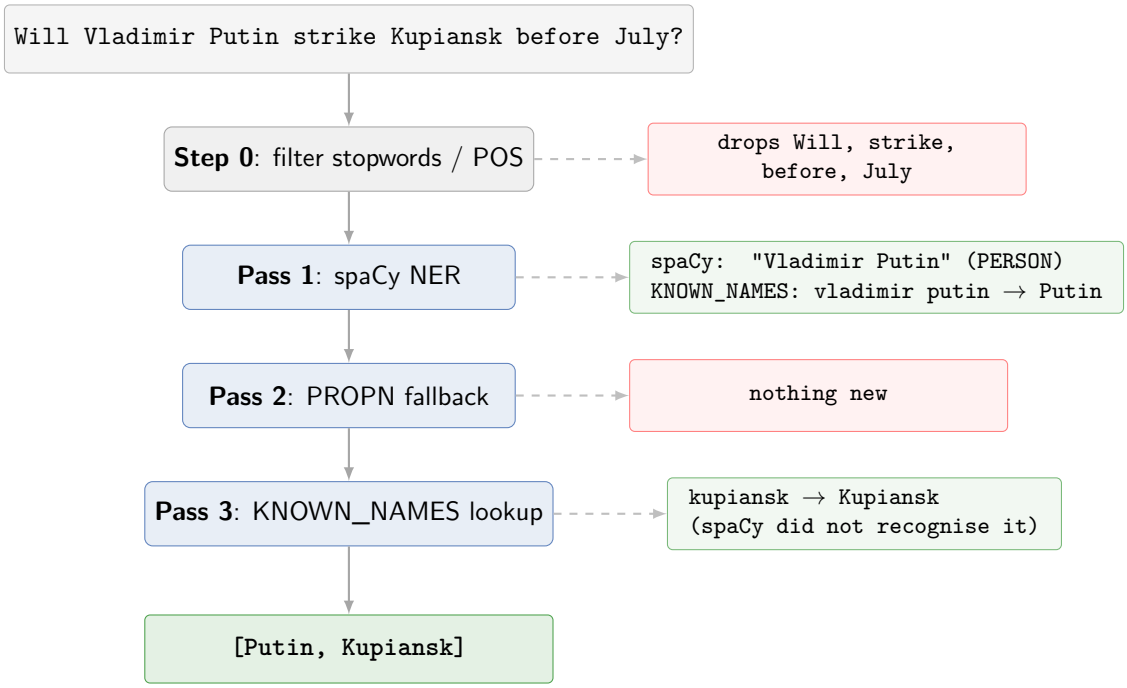

Figure 5: Worked example of the three-pass extractor on the question `"Will Vladimir Putin strike Kupiansk before July?"`. Step 0 drops stopwords. Pass 1 captures `"Vladimir Putin"` via spaCy NER, which `KNOWN_NAMES` then normalises to `Putin`. Pass 2 adds nothing. Pass 3 rescues `Kupiansk`, a Ukrainian frontline town that spaCy does not recognise.

Vocabulary

Two dictionaries in `pipeline_config.py` control the output of all three passes. Tables 9 and 10 summarise what each does.

Table 9: `STOPWORDS` categories (approx. 120 entries total).

Category	Examples
Market boilerplate	`yes`, `no`, `will`, `resolve`, `market`, `date`
Calendar tokens	months, `q1`–`q4`, `2023`–`2026`, `first`, `second`
Generic titles	`president`, `prime`, `minister`, `secretary`, `chancellor`
NER noise	`god`, `clock`, `trillion`, `emergency`, `victory`

Table 10: `KNOWN_NAMES` transformation categories (approx. 250 entries total).

Transformation	**Example mapping**
Name variants	`benjamin netanyahu` → `Netanyahu`
Demonyms	`israeli` → `Israel`
Abbreviations	`u.s.` → `USA`
Multi-word phrases	`gulf leaders summit` → `Gulf Leaders Summit`
Out-of-vocab names	`pokrovsk` → `Pokrovsk`
Possessive stripping	`Israel's` → `Israel`

Coverage

Running `ner_keywords.py` across all 1,801 resolved markets with trades yields 328 unique keywords, averaging 1.93 per market. The top 10 account for most of the coverage (Table 11), and a lot of keywords covers one or two markets each. The overall sentiment and financial coverage breakdown, including markets that produce no keywords at all, is reported in Section **??**.

Table 11: Top 10 keywords by market coverage.

Keyword	**Markets**
Israel	412
Trump	396
Iran	223
USA	182
Russia	167
Zelenskyy	116
Ukraine	105
Yemen	100
Putin	98
Gaza	67

1.2 Financial Times — ProQuest Corpus and FinBERT

The Financial Times track produces weekly sentiment scores for each market from FT coverage during the market's lifetime. This section runs in four stages: Corpus acquisition, parsing, FinBERT scoring, and weekly aggregation.

Corpus Acquisition

ProQuest does not expose a bulk download API at the UCD subscription tier — articles must be selected through the web interface and exported as `.txt` files. A market-by-market download is therefore not feasible. Queries were constructed manually, guided by the NER keyword distribution in Section **??**, with three targeting decisions per query.

Query scope. Bare single-keyword queries for common terms like `"Russia"` or `"Iran"` return tens of thousands of FT articles and cannot be practically exported. These were narrowed to two-keyword AND combinations (e.g. `"Russia" AND "Ukraine"`). For niche entities like `"Houthi"`, `"Kupiansk"`, or `"Milei"`, the article count is already bounded and single-keyword queries were used directly.

Date window. Each query was constrained to the lifetime of the markets it supports, with a 30-day lead added before the earliest market opens to capture the immediate pre-opening news cycle (Table 12).

Table 12: Date windows for three representative topics.

Topic	Window	Articles	Rationale
`russia_ukraine`	2023-01 – 2025-10	5,132	Continuous market coverage throughout the war.
`saudi_arabia_usa`	2025-04 – 2025-10	1,133	Markets opened spring 2025 around the Gulf Leaders Summit.
`elon_musk_trump`	2024-12 – 2025-10	2,040	Markets tracked a specific feud window.

Batching. ProQuest exports are capped at 500 articles per batch. High-volume topics required many batches: `russia_ukraine` in 53, `china_taiwan` in 35, while niche topics like `houthi` and `bolsonaro` required a single batch. Batch counts do not map linearly to article counts because some batch sizes are 100 others are 500.

In total, 72 topic corpora were acquired spanning 1 February 2023 to 31 October 2025. `parse_proquest.py` (Section **??**) produced 63,874 parsed articles; `finbert_scorer.py` (Section **??**) scored 51,647. The ∼19% gap consists of broad single-keyword corpora that were parsed early in development but later superseded by narrower two-keyword AND queries, and so were not carried through to scoring. Table 13 shows the 10 largest scored corpora.

Table 13: Top 10 ProQuest corpora by scored article count.

Topic	**Batches**	**Articles**	**Date range**
russia_ukraine	53	5,132	2024-08 – 2025-10
china_taiwan	35	3,390	2023-12 – 2025-10
moscow_ukraine	32	3,132	2024-02 – 2025-10
gaza_israel	28	2,668	2024-05 – 2025-10
hamas_israel	22	2,140	2024-06 – 2025-10
iran_usa	5	2,051	2023-11 – 2025-10
elon_musk_trump	5	2,040	2025-01 – 2025-10
iran_israel	4	1,757	2024-05 – 2025-10
trump_putin	19	1,648	2024-11 – 2025-10
netanyahu	3	1,472	2024-09 – 2025-10

References

[Polymarket(2025)] Polymarket. Polymarket developer documentation, 2025. URL https://docs.polymarket.com.

[Raasveldt and Mühleisen(2019)] Mark Raasveldt and Hannes Mühleisen. Duckdb: an embeddable analytical database. In *Proceedings of the 2019 ACM SIGMOD International Conference on Management of Data*, pages 1981–1984, 2019.

[warproxxx(2025)] warproxxx. poly_data: Polymarket data retriever, 2025. URL https://github.com/warproxxx/poly_data.

A Data Download

The raw snapshot is retrieved by running the following command from the project root:

```
curl -L -o data/raw/orderFilled_complete.csv.xz \
    https://polydata-archive.s3.us-east-1.amazonaws.com/orderFilled_complete.csv.xz
```

This downloads the 5.9 GB compressed archive directly from the `warproxxx/poly_data` S3 bucket. Full setup instructions are in the project README at johnstew324/polymarket-whale-watch.

B Pipeline Source Code